

Research article

Study on element mixing of titanium-stainless steel dissimilar metal welding pool and its effect on joint properties in nanosecond laser welding

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ABSTRACT

During the melting welding of titanium-stainless steel dissimilar metals, intermetallic compounds typically form within the weld seam, significantly compromising its mechanical properties. This study firstly used nanosecond pulse laser achieved effective welding of thin wall titanium and stainless steels by adjusting the mixing ratio of metal elements in the weld seam. A computational fluid dynamics (CFD) calculation model for thin-walled stainless steel titanium nanosecond pulse laser welding was established, and the physical mechanism of element mixing behavior was revealed. The effects of varying heat inputs on fluid flow, mixing of elements, and the evolution of microstructure in the weld pool were investigated. Regardless of whether the melting pool penetrates the titanium thin wall, titanium elements easily accumulate along the edges of the melting pool on the stainless steel side. When the melting pool fully penetrates the titanium sheet, the average titanium content in the weld seam exceeds 23 at%. Conversely, when the bottom of the melting pool is situated at the center of the titanium thin wall, the average titanium content within the weld seam is approximately 9.99 at%. No intermetallic compounds were found in the weld seam the strength of the weld can be increased about twice. Controlling the average content of Ti element in the weld seam to not exceed 10 at% is a key factor in obtaining excellent mechanical properties of welded joints, and it providing new welding strategies for titanium-stainless steel metal welding.

1. Introduction

The welded structures of titanium and stainless steel have broad potential applications in various fields, such as shipbuilding, petroleum, chemical engineering, aerospace, and medical devices [1–3]. However, this welding technology still faces significant challenges. Specifically, when different metal elements mix in the molten pool. Brittle intermetallic compounds frequently formed, which leading to a deterioration in the mechanical properties of the weld [4,5]. Revealing the effects of laser keyhole molten pool behavior on metal mixing, microstructure, and mechanical properties is essential for enhancing the strength of dissimilar metal joints.

It is well known that titanium and stainless steel welding has always been challenging. One of the main reasons is that the Fe-Ti binary alloy contains two types of intermetallic compounds, FeTi, which has a B2-type CsCl structure with a narrow compositional range of 47.5–50.3 at % Fe, and the hexagonal C14 MgZn₂-type Laves phase Fe₂Ti, which

contains 64.5–72.4 at% Fe [6]. These brittle intermetallic compounds lead to a significant reduction in the mechanical properties of the joint [7]. In addition, the physical properties of Ti and stainless steel, such as the coefficient of thermal expansion (Ti, 7.6 mm⁻¹K⁻¹, stainless steel, 17–18 mm⁻¹K⁻¹), resulting in excessive deformation and residual stress [8]. In order to solve the above problems, researchers primarily focus on adding additional alloying elements at the Ti-Fe interface to inhibit the formation of brittle intermetallic compounds. The selection strategy for alloying elements can be divided into two categories. The first category includes alloying elements that do not form brittle intermetallic compounds with Ti, such as V, Nb, Ta, Mg, W, Zr, and Mo. The second category consists of alloying elements that do not form brittle intermetallic compounds with Fe, such as Cu, Ni, Ag, Co, and Mn [9]. Unfortunately, there is no alloying element that does not produce an intermetallic compound with Fe or a brittle intermetallic compound with Ti. Therefore, researchers typically choose one or several of these two types of elements respectively for combined addition, such as Cu/V

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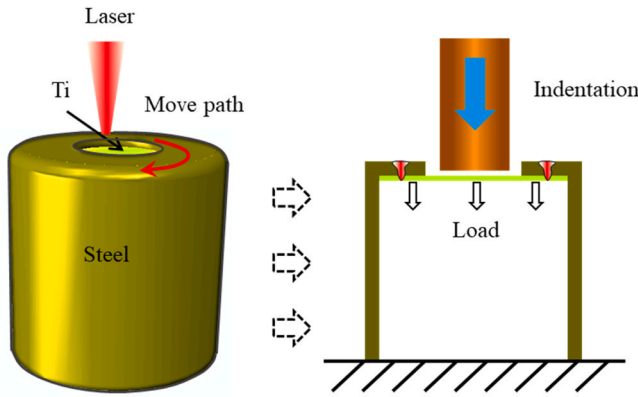


Fig. 1. Schematic diagram of the laser welding structure of titanium-steel dissimilar metal lap joints and mechanical performance testing.

Table 1
Welding parameters.

Number	Power (kW)	Velocity (mm/s)	Frequency (kHz)	Pulse width (s)	Defocus (mm)
1	0.16	110	CW	0.000002	+ 3
2	0.18	110	CW	0.000002	+ 3
3	0.2	110	125	0.000002	+ 3
4	0.2	110	135	0.000002	+ 3
5	0.2	110	CW	0.000002	+ 3

[10], Cu/Nb [11], and Ni/Nb [12]. Additionally, some scholars have explored adding medium entropy or high entropy alloys containing Fe or Ti elements at the steel/titanium interface, aiming to form high entropy alloys in the weld instead of brittle intermetallic compounds [13–15]. Unfortunately, intermetallic compounds such as FeTi or Fe₂Ti still form in the weld seam. Moreover, other researchers [16] first created Ti/steel composite layers through explosive welding, placing them between the stainless steel and titanium plates to be welded. Then laser-welded the steel/steel and Ti/Ti interfaces separately, avoiding direct mixing of Ti and steel. However, this welding process increases the complexity of the welding procedure, raises production costs, and can present considerable challenges when the wall thickness of the welded components is very thin.

Wang et al. [17] and Chen et al. [18] attempted to weld titanium and stainless steel without adding an intermediate layer. When the laser beam was positioned at the center of the interface, a continuous distribution of a large number of Ti-Fe intermetallic compounds within the weld seam led to significant brittleness, resulting in fracture. However, when the laser beam was shifted 0.6 mm towards the stainless steel side, a relatively small and discontinuous region of Ti-Fe intermetallic

compounds formed in the weld seam due to the reduction of molten titanium alloy, and the joint tensile strength achieved 150 MPa. Nevertheless, they did not specify the optimal range of Ti or Fe content in the weld seam for achieving the best welding performance between titanium and stainless steel. This indicates that precisely controlling the ratio of Ti and Fe elements in the molten pool might be an effective and practical approach, particularly for structures like thin-walled titanium and steel where adding an intermediate layer is challenging.

According to the Ti-Fe binary phase diagram, the maximum solubility of Ti in α -Fe is approximately 10 at%. If the ratio of Fe to Ti in the molten pool is accurately controlled to not exceed 9:1, and the cooling rate is fast during the welding process, the supersaturated α -Fe can be retained at room temperature. Thereby preventing the formation of Fe₂Ti intermetallic compounds. Traditional pulsed lasers or continuous fiber lasers face challenges such as penetrations, large deformations, and low welding strength when welding dissimilar metal thin sheets (thickness less than 0.2 mm) [19]. The nanosecond pulsed laser welding technology, as a new welding technique, allows for precise control of the welding heat input, providing significant advantages in the connection of ultra-thin dissimilar metal sheets [20]. To accurately control the mixing ratio of Fe and Ti elements in the molten pool and achieve a faster cooling rate, nanosecond pulsed lasers are used as the welding heat source.

There have been many studies on welding dissimilar metals using pulsed lasers, but most of them use laser heat sources with pulse widths in the millisecond range. Nanosecond pulse laser heat source has high peak energy and short pulse width, which has high advantages in precise adjustment of heat input. In recent years, researchers have gradually started using nanosecond pulse lasers to weld thin-walled lap joints such as Steel-Cu [21] and Al-Cu [22,23], achieving good welding results, especially in suppressing the formation of intermetallic compounds. Kumar et al. [24] found in nanosecond pulsed laser welding of Al-Cu that the heat input was reduced from 3.8 J/mm to 2.3 J/mm, and the thickness of intermetallic compounds was reduced from 5 μ m to 1 μ m. In addition, Zhemchuzhnikova et al. [25] found that cracking can be

Table 2
Physical parameters of stainless steel alloy [40,41] and pure titanium [42–44].

Property	316 L	Ti
Density, ρ (kg/m ³)	7249	4400
Thermal conductivity of solid metals, k_s (W/m/K)	19.2	16.0
Thermal conductivity of liquid metal, k_L (W/m/K)	22.0	12.0
Specific heat, C (J/kg/K)	755	570
Dynamic viscosity, μ (kg/m/s)	0.006	0.005
Latent heat of fusion, h (J/kg)	2.47×10^5	2.86×10^5
Surface tension, σ (N/m)	1.7	1.68
Surface tension gradient, A (N m ⁻¹ K ⁻¹)	-4.3×10^{-4}	-2.4×10^{-4}
Diffusion coefficient, D	2.765	7.997
Melting point, T_m (K)	1697	1928

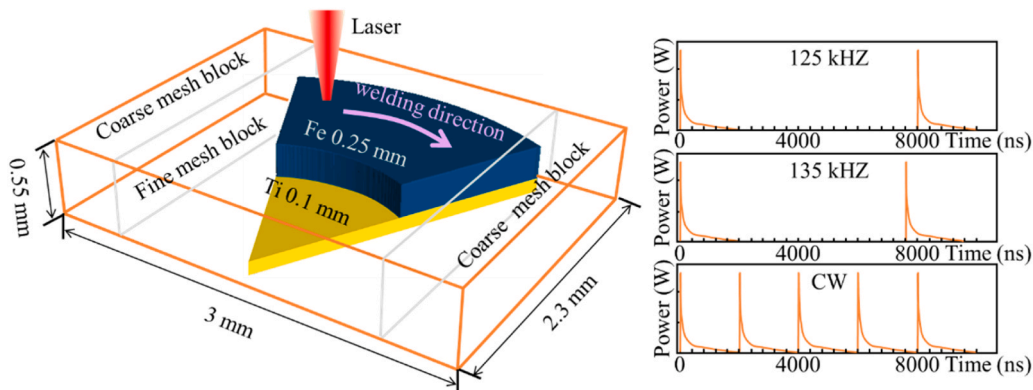


Fig. 2. Model setup and pulse waveform shape.

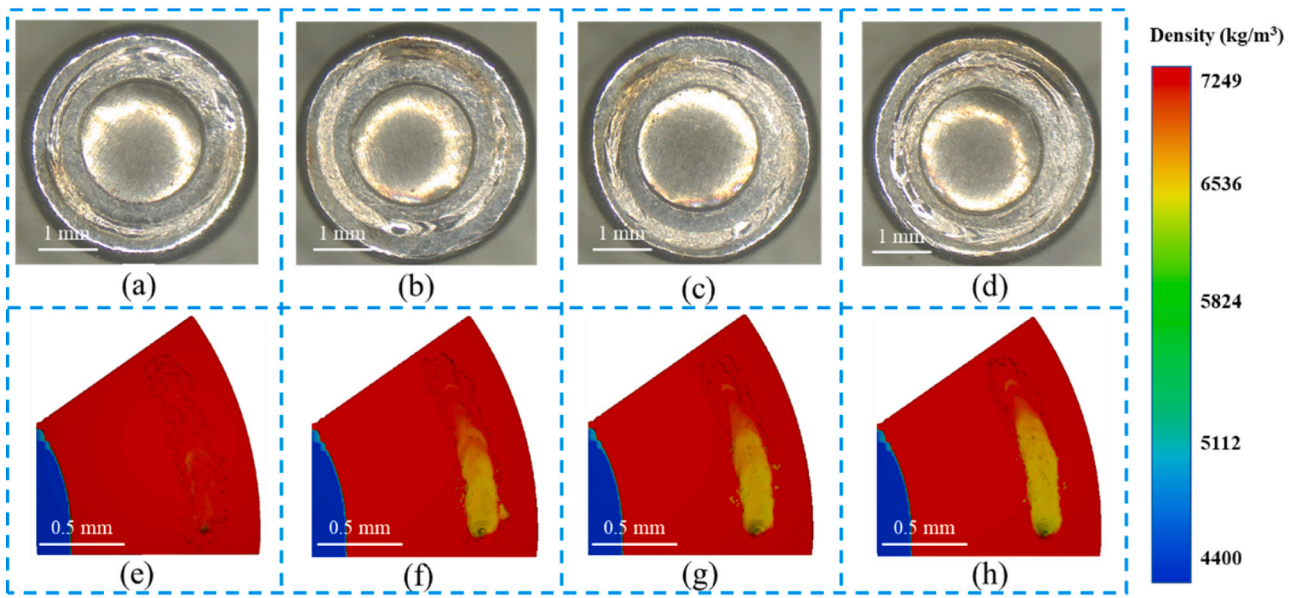


Fig. 3. Appearance of weld seams produced by different process parameters. (a-d) the appearance of weld seams obtained by process parameters 1#, 3#, 4# and 5#, respectively. (e-h) the appearances of the weld seams obtained from simulation for process parameters 1#, 3#, 4# and 5#, respectively.

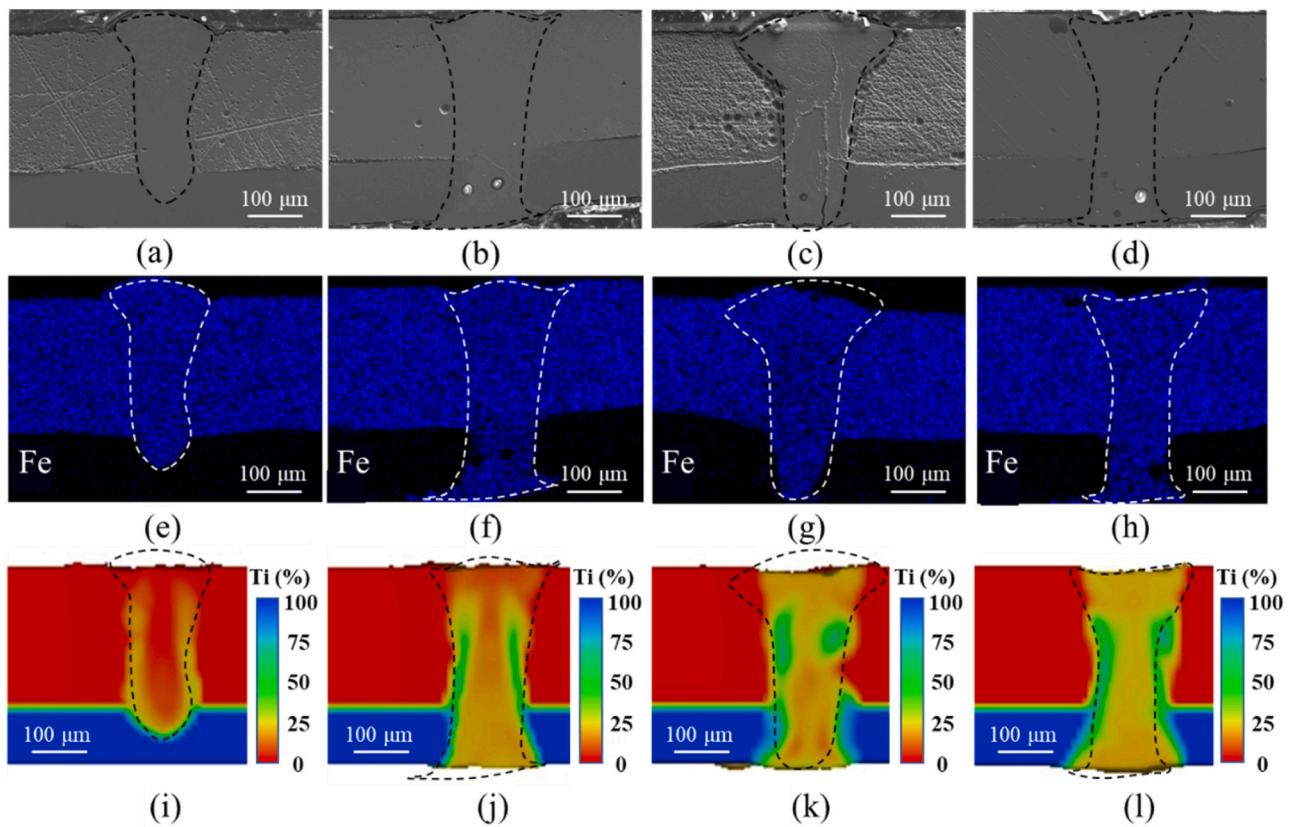


Fig. 4. Comparison of experimental and simulation results on the cross-sectional morphology and element concentration field of weld seams. (a-d) the weld seam cross-sectional profiles produced by parameters 1#, 3#, 4# and 5#, respectively. (e-h) the energy spectrum mapping of the Fe element in the weld seam cross-sections for process parameters 1#, 3#, 4# and 5#, respectively. (i-l) the Ti element distribution image obtained from simulation for process parameters 1#, 3#, 4# and 5#, respectively.

completely avoided when welding 6061 aluminum alloy with nanosecond pulse laser. However, so far, there have been no reports of using nanosecond pulse lasers to weld thin-walled titanium and stainless steels.

Dynamic detection of metal mixing behavior during the laser

welding process presents significant challenges. With the rapid development of computational fluid dynamics (CFD), researchers have recently begun to apply CFD techniques to assist in studying the metal mixing process in the laser welding of dissimilar metals. Utilizing CFD models allows for the identification of interrelated reactions among heat

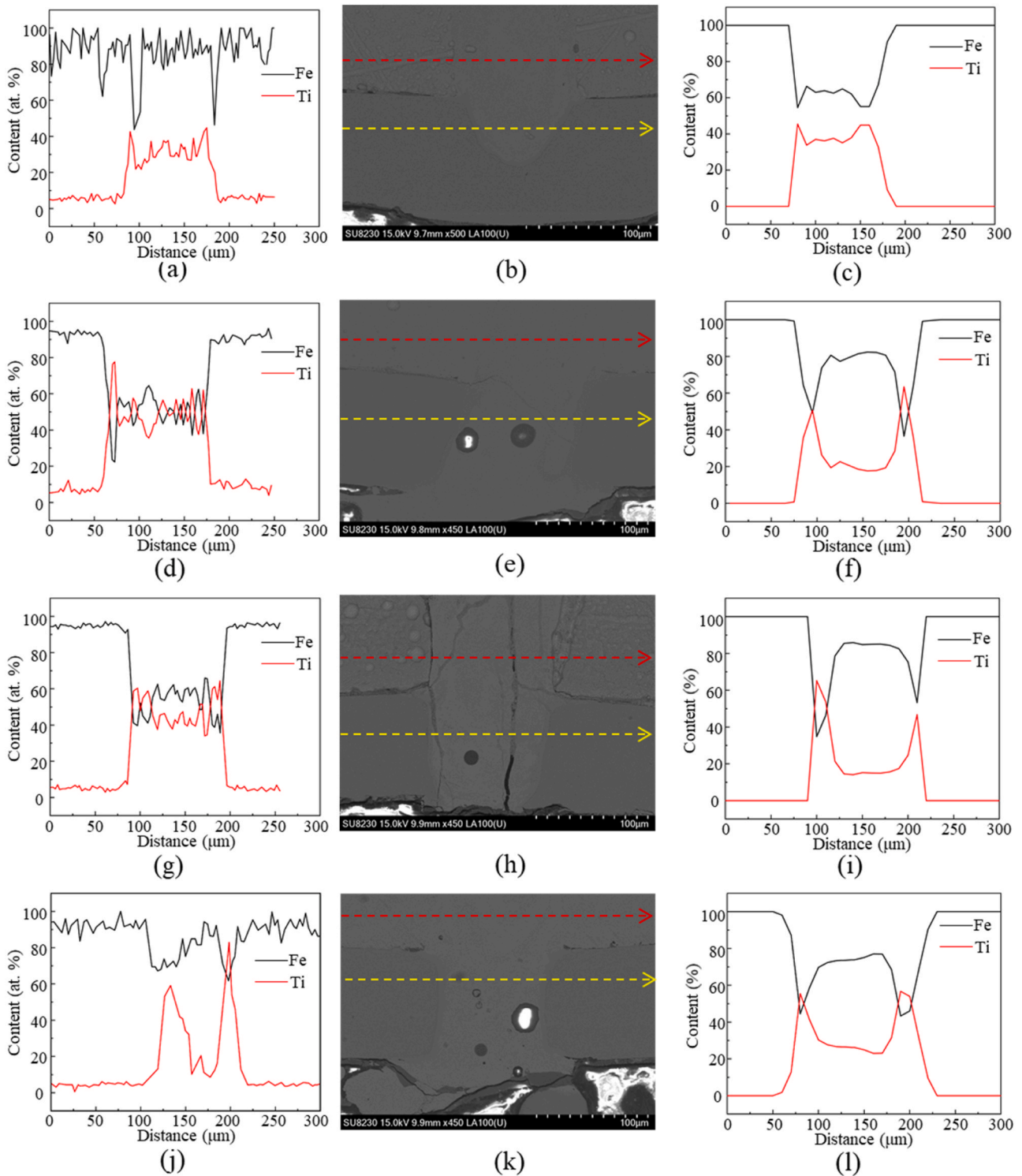


Fig. 5. Comparison of the distribution characteristics of Fe and Ti elements on the stainless steel side. (a), (d), (g) and (j) are the elements distribution curves obtained from line scanning energy spectrum for process parameters 1#, 3#, 4# and 5#, respectively. (b), (e), (h), and (k) illustrate the energy spectrum line-scanning paths across distinct weld seam cross-sections. The red dashed line denotes the scanning path on the stainless steel side, while the yellow dashed line corresponds to the pure titanium side. (c), (f), (i) and (l) are the elements distribution curves obtained from simulation for process parameters 1#, 3#, 4# and 5#, respectively.

flux, temperature distributions, driving forces, fluid flow, and metal amalgamation within the molten pool. Huang et al. [26] predicted the heat transfer, fluid flow, and metal mixing behavior in the molten pool during the Al-Cu dissimilar metal laser welding process using a CFD model, uncovering the fundamental physical principles of the metal

mixing process. They conducted a quantitative assessment of how various welding parameters impact laser absorption, temperature, driving forces, fluid movement, and the spatial arrangement of metal mixing within the molten pool. Additionally, they [27] studied the influence of oscillating lasers on molten pool flow using the developed

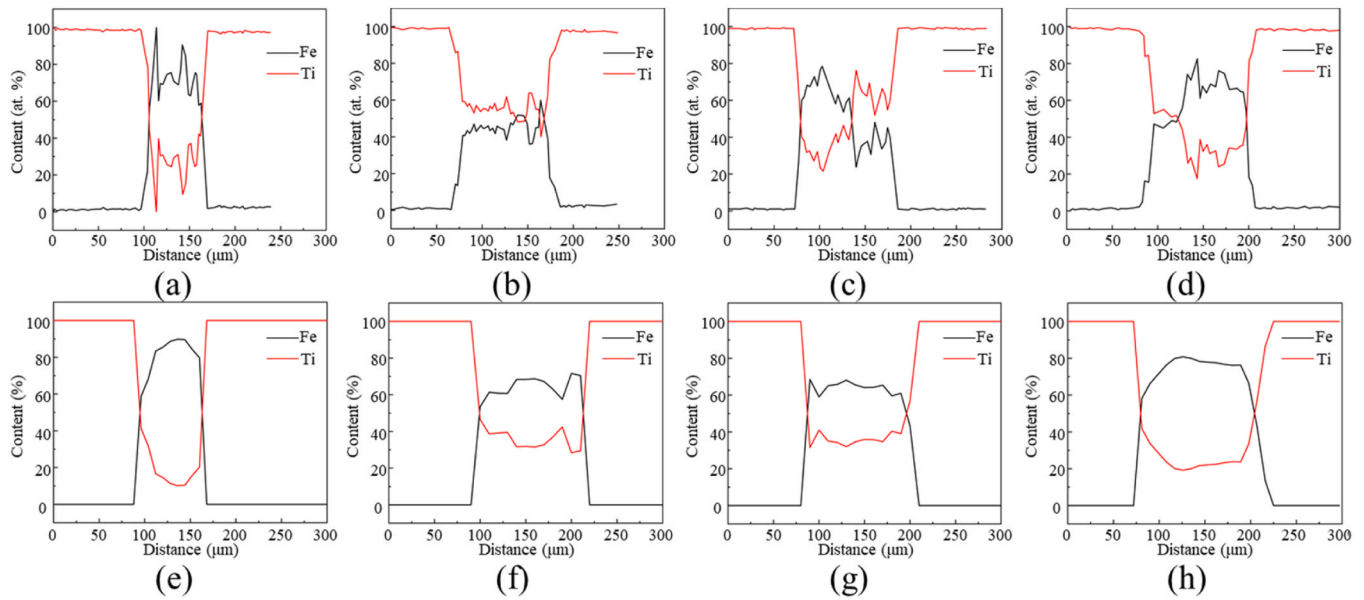


Fig. 6. Comparison of the distribution characteristics of Fe and Ti elements on the titanium side. (a-d) the element distribution curves obtained from line scanning energy spectrum for process parameters 1#, 3#, 4# and 5#, respectively. (e-h) the element distribution curves obtained from simulation for process parameters 1#, 3#, 4# and 5#, respectively.

CFD model and discovered that four distinct fluid flow patterns determine the mixing state of Al and Cu elements in the molten pool. In addition, Raza et al. used CFD simulations to study the dynamic changes in the molten pool during Al Cu welding. Based on the CFD model, the optimal process parameters for pulsed laser welding (PLW) [28] and continuous laser welding (CW) [29] were optimized, providing valuable information for the design and optimization of Al Cu lap joints. There are few reports on studying the flow behavior of Ti Steel laser welding melt pool through CFD models. Isaev et al. [30] established a 3D numerical model for laser welding of titanium and steel, in which the fluid flow and metal mixing were not explicitly calculated. The possible positions of metal melting and mixing were inferred solely based on the temperature field. Wu et al. [31] created a three-dimensional model for laser welding of steel and titanium that simulated heat transfer between the two dissimilar metals, along with fluid movement, keyhole dynamics, and element diffusion. However, this model overlooked the influence of fluid flow on the mixing behavior of dissimilar metals, with mixing relying solely on diffusion. Consequently, at the interface between the dissimilar metals, only a diffusion layer is observable instead of a distinct mixing layer. So far, research on CFD simulation of dissimilar metal laser welding has focused on the laser pulse width in the millisecond range, and there have been no reports on CFD models with laser pulse width in the nanosecond range. Furthermore, there is no model that reveals the impact of changes in melt density on melt pool dynamics. Experiments can elucidate the influence of different factors on metal mixing. Nonetheless, the underlying physics that drives the mixing process remains inadequately understood. All theoretical explanations regarding the dynamic mixing phenomenon of elements during welding are typically based on extrapolations from the microstructural characterization of the joints. Experimental measurement of metal dynamic mixing during welding remains challenging, CFD simulations offer an alternative approach to help understand this important issue.

In summary, the difficulty of direct welding between Steel-Ti lies in the high propensity for the formation of intermetallic compounds, which can lead to joint failure. Nanosecond pulsed lasers can effectively suppress the formation of intermetallic compounds during welding of dissimilar metals such as Al-Cu. However, it remains unclear whether nanosecond pulsed lasers are effective for welding such as Steel-Ti, and further verification is needed. In addition, existing CFD computational

models for pulsed laser welding of dissimilar metals focus only on laser heat sources with millisecond-level pulse widths, and there have been no reports on modeling and analysis of nanosecond pulsed laser heat sources with high peak energy and low pulse width. To gain a deeper understanding of the physical mechanism of nanosecond pulsed laser welding of Steel-Ti dissimilar metal joints, it is necessary to establish a CFD model to study its influence on the mixing behavior of dissimilar elements.

In this study, a CFD model for nanosecond laser welding of thin-walled titanium-stainless steel dissimilar metal lap joints was developed in conjunction with experimental investigations. A quantitative analysis was conducted to assess how various welding parameters influence laser absorption, temperature, driving forces, fluid flow, and the spatial distribution of metal mixing within the molten pool. The research results indicate that when the average content of Ti element is 9.99 at%, there are no cracks or intermetallic compounds produced in the weld, and the strength of the weld can be increased by about twice. This study not only verifies the feasibility of nanosecond pulse laser welding thin-walled titanium-stainless steel, but also offers theoretical support and practical guidance for future optimization of titanium-stainless steel metal welding.

2. Experimental materials and methods

This study employed a combined approach of experimentation and simulation. Laser welding experiments were conducted to determine the effects of different welding parameters on the microstructure and mechanical properties. Subsequently, a CFD model was developed to simulate the fluid flow and metal mixing during the welding process.

2.1. Experimental

In order to facilitate the mechanical performance testing of dissimilar metal lap joints, a titanium-steel dissimilar metal welding structure was designed, as shown in Fig. 1. A 0.1 mm commercial pure titanium sheet was placed at the bottom of a 0.25 mm stainless steel casing. The diameter of the stainless steel casing is 4 mm, and its height is 4 mm. The welding heat source was a Trumpf TruPulse nano laser with a wavelength of 1059 nm, a maximum average power of 200 W, and a

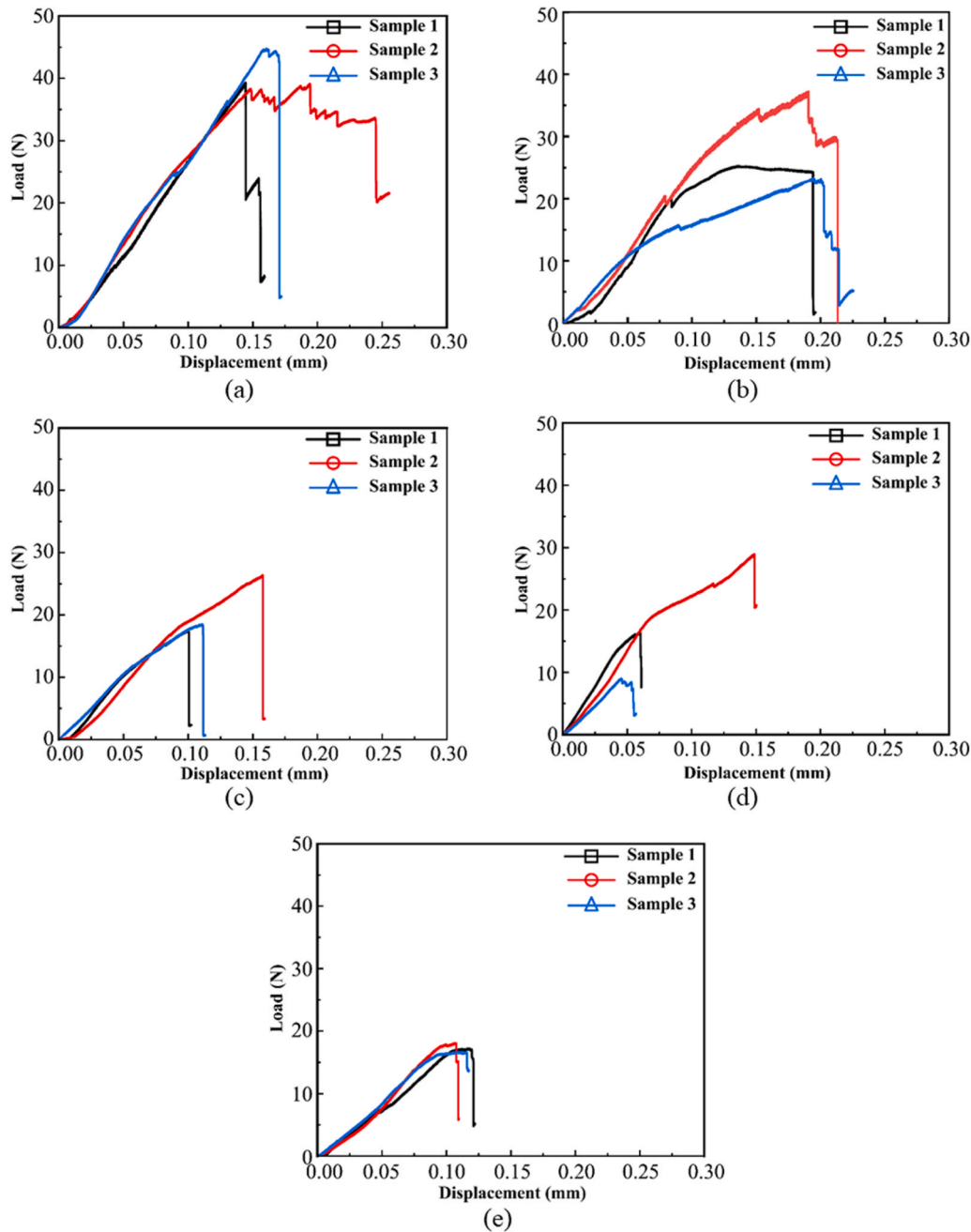


Fig. 7. Load-displacement curves of welded joints produced by different welding parameters. (a-d) the joints obtained from process parameters 1# - 5#, respectively.

maximum peak power of 10000 W. A circular trajectory was selected for the laser movement during the welding process.

To examine the impact of various heat inputs on metal mixing, microstructure, and the mechanical properties of the weld, four distinct welding parameters were designed, as detailed in Table 1. Compare the parameters of 1#, 2# and 5#, the influence of laser power on Ti element distribution and joint strength can be studied. Compare the parameters of 3#, 4# and 5#, We hope to slightly adjust the heat input by controlling the frequency, thereby playing a role in fine-tuning the Ti element content. Four samples were welded for each experimental parameter, three of which were used for mechanical property testing. The testing method is shown in Fig. 1, one sample was used to observe the cross-sectional profile and microstructure of the weld. Before observation, the entire welded component was cut from the center. Grinding begins with sandpaper featuring a roughness of 80 mesh, and the roughness is progressively reduced until finishing with 5000 mesh

sandpaper. Finally, a silica suspension is applied to complete the polishing process. The metallographic sample surface was then etched with 4 % nitric acid alcohol for 15 seconds, followed by an observation of the cross-sectional morphology of the weld.

2.2. CFD model

A 3D numerical simulation model was created utilizing Flow-3D software. The simulation domain is shown in Fig. 2, it represents just a fraction of the full welded structure, roughly 1/12 of the whole assembly. The dimensions of the computational domain are $3 \times 2.3 \times 0.55$ mm, and it has been divided into a transitional mesh, with a minimum mesh size of 10 μ m and a maximum mesh size of 100 μ m, and the total number of elements is 1667305. The boundary conditions applied to the six surfaces of the computation domain are uniform pressure (atmospheric pressure) boundary conditions. The

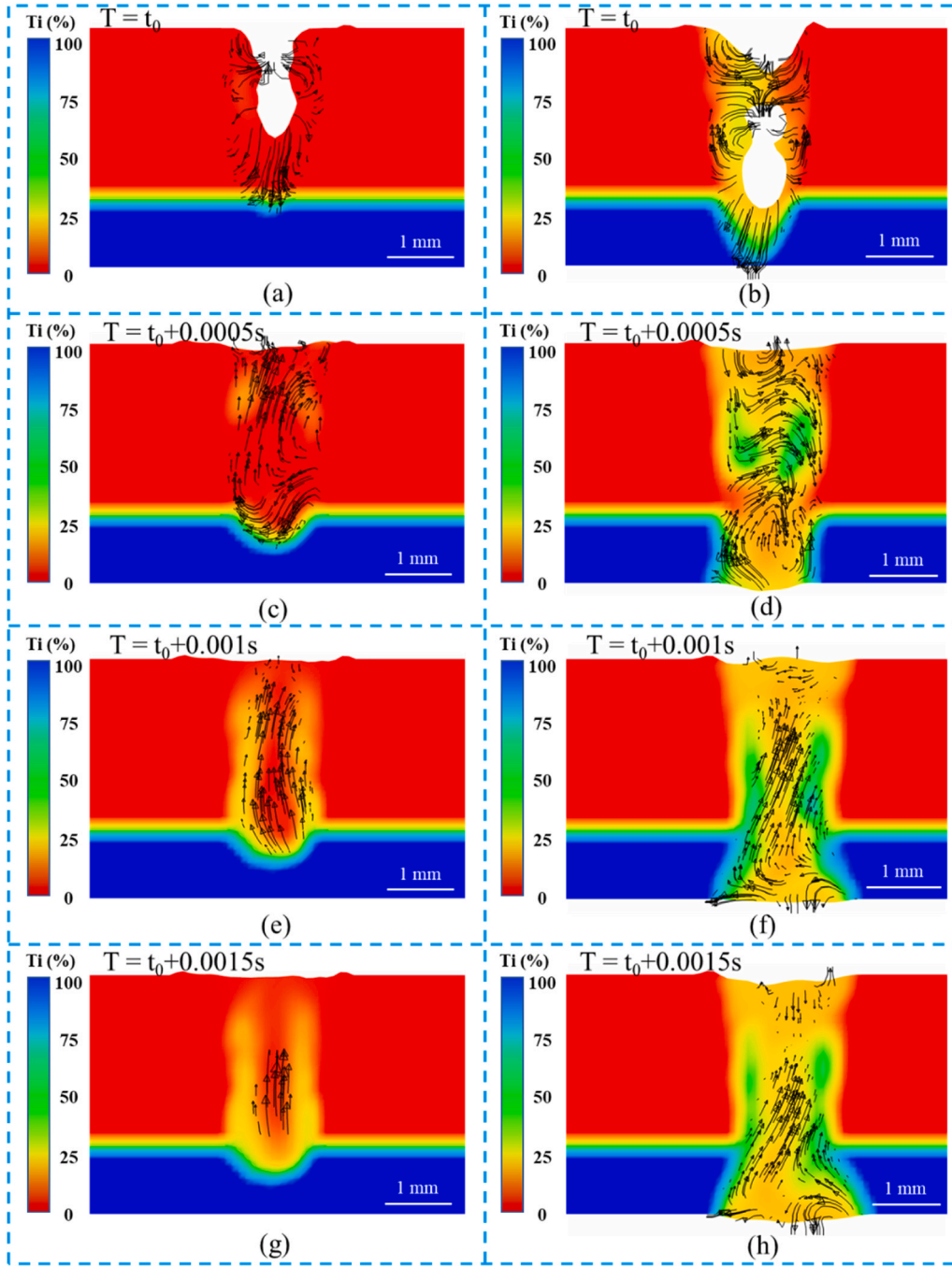


Fig. 8. The fluid flow trace and element distribution at different times of the weld cross section from the simulation results. (a-d) simulation results for 1# process parameters, (e)-(f) simulation results for 5# process parameters.

model uses welding parameters (laser power, pulse frequency, defocus amount, and welding speed) and material properties as input to predict the temporal and spatial variations of temperature, fluid flow, and element distribution in the molten pool. The pulse laser waveforms at different frequencies are shown in Fig. 2, CW means there is no time interval between each laser pulse. The maximum incremental time step during model computation is 1×10^{-8} s. All boundary conditions are pressure boundary conditions.

2.2.1. Governing equation

The temperature field, flow field and element mixing of molten pool metal in the model are calculated by the following governing equations, where the continuum equation is as follows [32]:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \quad (1)$$

Where, ρ is to calculate the density of the fluid in the domain grid cell, and $\mathbf{v}$ is the flow velocity of the liquid metal fluid. The momentum conservation equation is as follows [33]:

$$\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} = -\frac{1}{\rho} \nabla P + \mu \nabla^2 \mathbf{v} + \mathbf{G} \quad (2)$$

Where, μ is fluid viscosity, P is fluid pressure, and $\mathbf{G}$ is physical acceleration. The energy equation based on enthalpy and continuity formula is described as follows [34]:

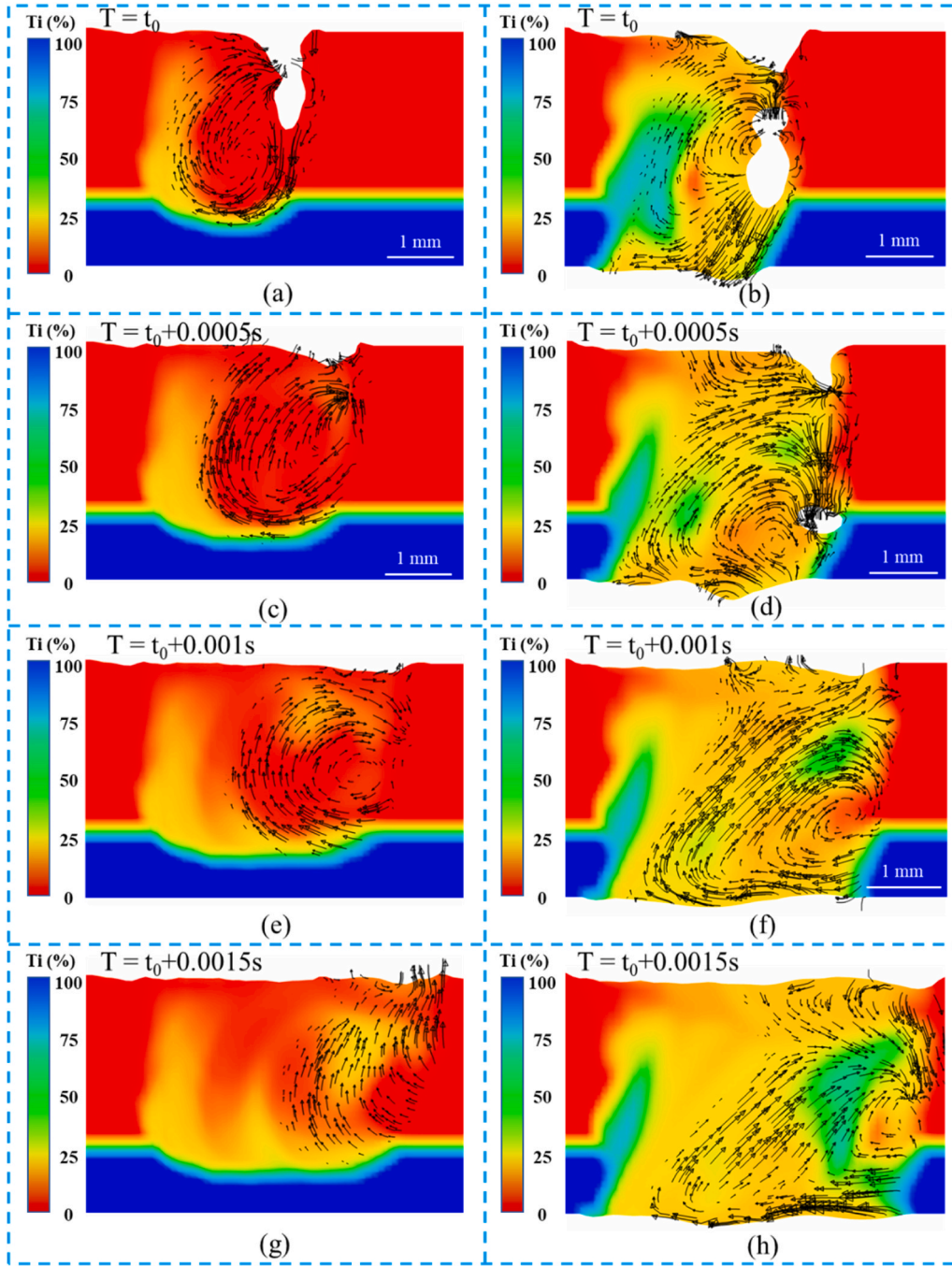


Fig. 9. The fluid flow trace and element distribution at different times of the weld longitudinal section from the simulation results. (a-d) simulation results for 1# process parameters, (e)-(f) simulation results for 5# process parameters.

$$\frac{\partial h}{\partial t} + \nu \cdot \nabla h = \frac{1}{\rho} \nabla \cdot (k \cdot \nabla T) P \quad (3)$$

$$h = C_p \cdot T + (T_L - T_S) \cdot L_f \quad (4)$$

Where h is the enthalpy of heat, k is the thermal conductivity, C_p is the specific heat capacity, L_f is the enthalpy of melting, T_S is the solid temperature, and T_L is the melt temperature. The equation for the conservation of matter is as follows:

$$\frac{\partial X}{\partial t} + \nu \cdot \nabla X = \nabla \cdot (D \cdot \nabla X) P \quad (5)$$

Where X is the concentration of Ti element and D is the diffusion

coefficient of the element.

2.2.2. Physical model

During the laser keyhole welding process, the laser undergoes multiple reflections, which can be simulated using the ray tracing method to model this physical phenomenon [35]. This method divides the laser beam into multiple rays. For each ray, the reflection path is tracked using the law of reflection, and the laser power absorbed by the keyhole is calculated using the Fresnel equations [36]. In the simulation process, the laser heat source is typically treated with surface heat flux boundary conditions, with the Gaussian function being widely accepted. The laser heat source model is shown in Eq. (6):

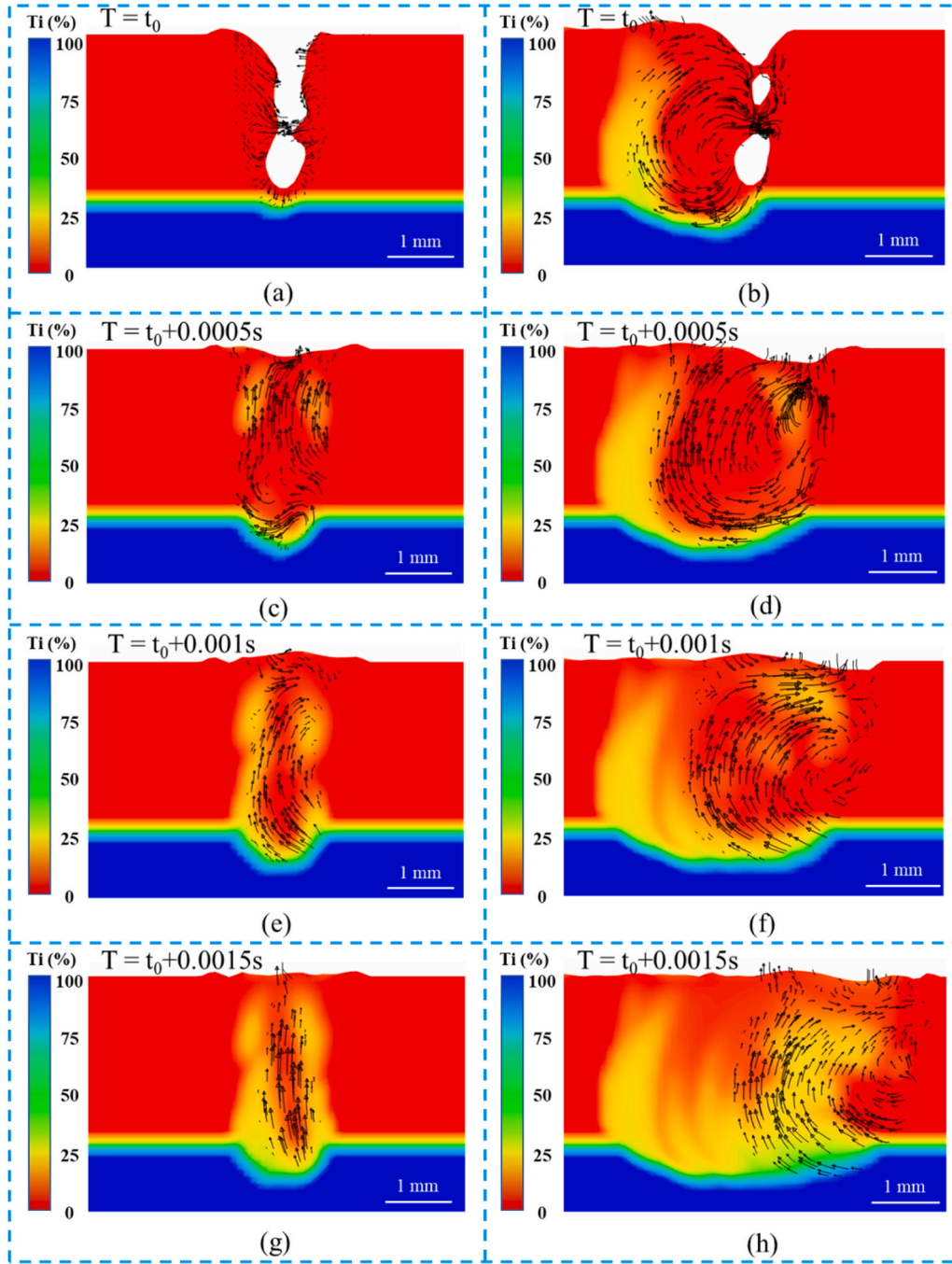


Fig. 10. The snapshot of the 1# process parameter simulation is taken by slicing along the cross-section and longitudinal section of the weld seam, when the densities of the two alloys are the same. (a-d) represent the cross-sectional slices, (e-f) represent the longitudinal slices.

$$Q(x, y, z) = \frac{2P}{2\pi r_0^2} \cdot e^{-\left(\frac{x^2 + y^2}{r_0^2}\right)} \quad (6)$$

Where P is the laser power and r_0 is the spot diameter. Surface heat flux Q makes metal melt and evaporate, and the reaction force acting on the surface of molten pool after melt evaporation is called recoil pressure. The recoil pressure model used in this study is derived from Clausius-Clapeyron equation, and its form is as follows [37]:

$$P(T) \cong 0.54P_{\text{sat}}(T) = 0.54 \cdot P_0 \cdot e^{\left(\frac{L_v(T - T_b)}{RT_b}\right)} \quad (7)$$

Where, 0.54 is a factor that depends on environmental pressure, P_{sat} is

the saturation vapor pressure, P_0 is the standard atmospheric pressure, and L_v is the latent heat of evaporation. T , T_b , and R represent the surface temperature, boiling point temperature, and universal gas constant, respectively. The driving forces of fluid flow include not only recoil pressure but also surface tension and buoyancy. The flow generated on the surface of molten metal under the influence of surface tension is referred to as Marangoni flow. In this study, surface tension is related to the concentration of Ti elements, which can be expressed as Eq. (8) [38]:

$$\sigma = \sigma_{\text{Ti}} \cdot X + \sigma_s \cdot (1 - X) \quad (8)$$

Where σ_{Ti} is the surface tension of titanium melt and σ_s is the surface tension of stainless steel melt. The buoyancy model is represented by Boussinesq as Eq. (9) [39]:

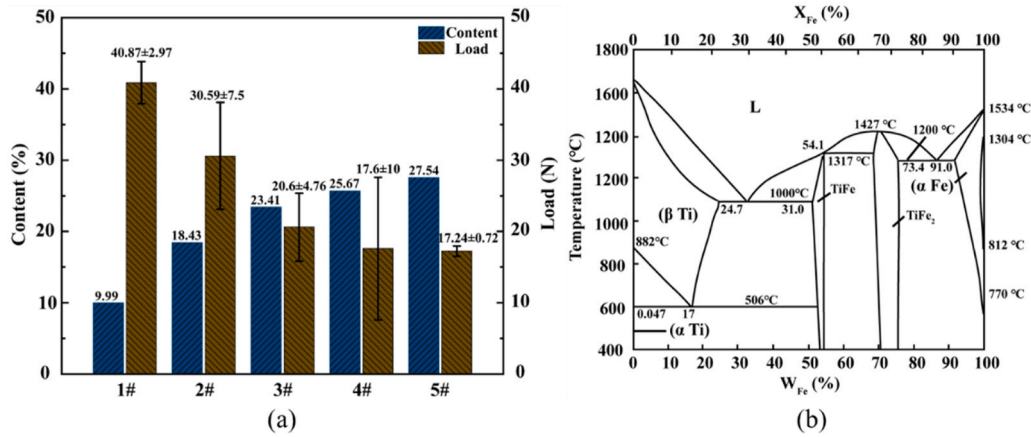


Fig. 11. Failure load corresponding to different Ti content and Ti-Fe phase diagram. (a) the Ti element content in the weld obtained by different welding parameters and the maximum load when the weld fails, error bars is STDEV, (b) Ti-Fe phase diagram [45].

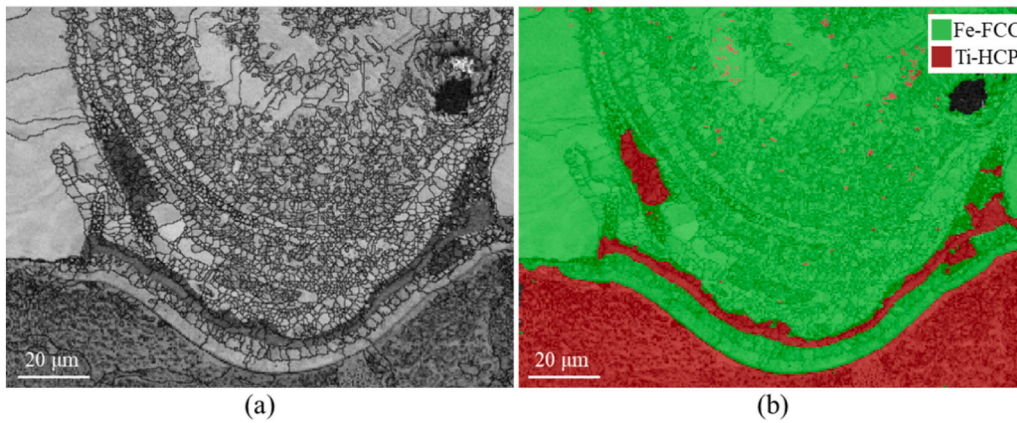


Fig. 12. Electron Back-Scattered Diffraction (EBSD) image of the welding seam cross-section produced by 1# parameter. (a) band contrast (BC) image, (b) phase distribution image.

$$F_b = \rho \cdot g \cdot \beta \cdot (T - T_0) \quad (9)$$

Where, g is gravity, ρ is fluid density, β is thermal expansion coefficient, ρ and β are related to the concentration of Ti element. Assuming the initial temperature of the material and the environment is 298 K, the thermophysical parameters of stainless steel and pure titanium are shown in Table 2.

3. Result

3.1. Weld profile comparison

The appearance of the weld seams obtained under four different welding parameter conditions is shown in Fig. 3. Specifically, Fig. 3a-d depict the appearance of weld seams obtained by process parameters 1#, 3#, 4# and 5#, respectively. While Fig. 3e-h illustrate the appearances of the weld seams obtained from simulation for process parameters 1#, 3#, 4# and 5#, respectively. From both experimental and simulation results, it can be observed that changing the welding frequency has little effect on the appearance of the weld seam. Furthermore, the simulation results indicate that as the overall heat input increases, the mixing degree of the Ti element gradually increases.

The welds obtained by different welding parameter were cut respectively, and the obtained weld cross section morphology and element distribution were shown in Fig. 4. Fig. 4a-d depict the weld seam cross-sectional profiles produced by parameters 1#, 3#, 4# and

5#, respectively. The black dashed line representing the outline of the weld cross-section. Fig. 4e-h show the energy spectrum mapping of the Fe element in the weld seam cross-sections for process parameters 1#, 3#, 4# and 5#, respectively. The white dashed line indicating the weld cross-section outline. Fig. 4i-l illustrate the Ti element distribution image obtained from simulation for process parameters 1#, 3#, 4# and 5#, respectively. The black dashed line representing the weld cross-section outline.

In the weld seam obtained from the 1# process parameters, when the molten pool was stable, the edge of the molten pool on the titanium side was located at the center of the titanium thin wall, without penetrating through it. For the weld seams from 3#-5# process parameters, when the molten pool was stable, the molten depth of the weld fully penetrated the titanium thin wall. Comparison of the experimental and simulation results indicates that the profile morphology of the weld obtained from experiments is quite similar to that obtained from simulations. The simulation results clearly show that there are significant Ti element enrichment zones on both sides of the weld profile on the stainless steel side. To further verify this phenomenon experimentally, EDS line scanning analysis was conducted on the stainless steel side and the pure titanium side of the dissimilar metal weld joint.

3.2. Element distribution on weld cross section

In the weld cross section obtained by different welding parameters, the distribution characteristics of Fe and Ti elements on the stainless steel side are shown in Fig. 5. Fig. 5a,d,g,j display the elements

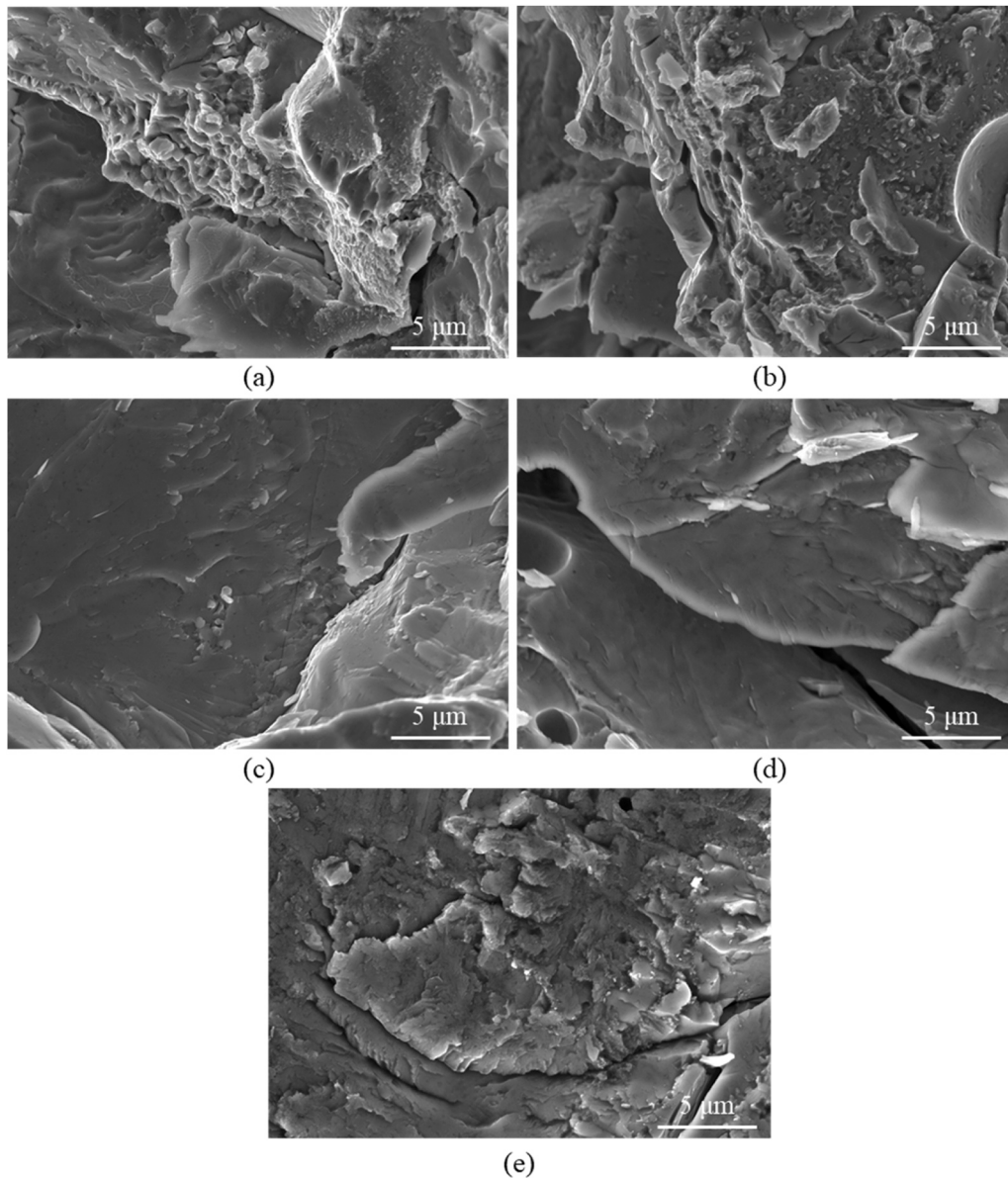


Fig. 13. Comparison of fracture morphology. (a) 1# process parameters, (b) 2# process parameters, (c) 3# process parameters, (c) 4# process parameters, (d) 5# process parameters.

distribution curves obtained from line scanning energy spectrum for process parameters 1#, 3#, 4# and 5#, respectively. Fig. 5b,e,h,k show the EDS line scanning paths in the different weld seam cross-sections, with the red dashed line indicating the line scanning path on the stainless steel side and the yellow dashed line indicating the path on the pure titanium side. Fig. 5c,f,i,l present the elements distribution curves obtained from simulation for process parameters 1#, 3#, 4# and 5#, respectively. It can be conclude that, regardless of whether the results are obtained from experiments or simulations of dissimilar material welding joints, there are Ti element enrichment zones on both sides of the weld profile on the stainless steel side. When welding with process parameter 1#, the enrichment of Ti elements is the smallest. The elemental distribution characteristics obtained from simulations and experiments show a highly similarity.

Fig. 6 compares the distribution characteristics of Fe and Ti elements on the pure titanium side of the weld cross-section obtained from different welding parameters. Fig. 6a-d show the element distribution curves obtained from line scanning energy spectrum for process parameters 1#, 3#, 4# and 5#, respectively. The line scan paths are

indicated by the yellow dashed lines in Fig. 5b,e,h,k, respectively. Fig. 6e-h present the element distribution curves obtained from simulation for process parameters 1#, 3#, 4# and 5#, respectively. When welding with 1# process parameter, the content of Fe element in the weld on the pure titanium side is significantly higher than that of Ti element, followed by 4# process parameter. In contrast, for welding with process parameters 2# and 3#, the content of Fe elements on the pure titanium side is close to that of Ti elements, indicating a more uniform mixing of Fe and Ti elements at the bottom of the molten pool under these two welding conditions.

3.3. Joint strength

Fig. 7 shows the load-displacement curves of welded joints produced by different welding parameters. Fig. 7a-e represent the joints obtained from welding with parameters 1#-5#, respectively. The average load at which the joint welded with parameter 1# fails in tension is approximately 40.87 N. The average load for the joint welded with parameter 2# is approximately 30.59 N. The average load for the joint welded with

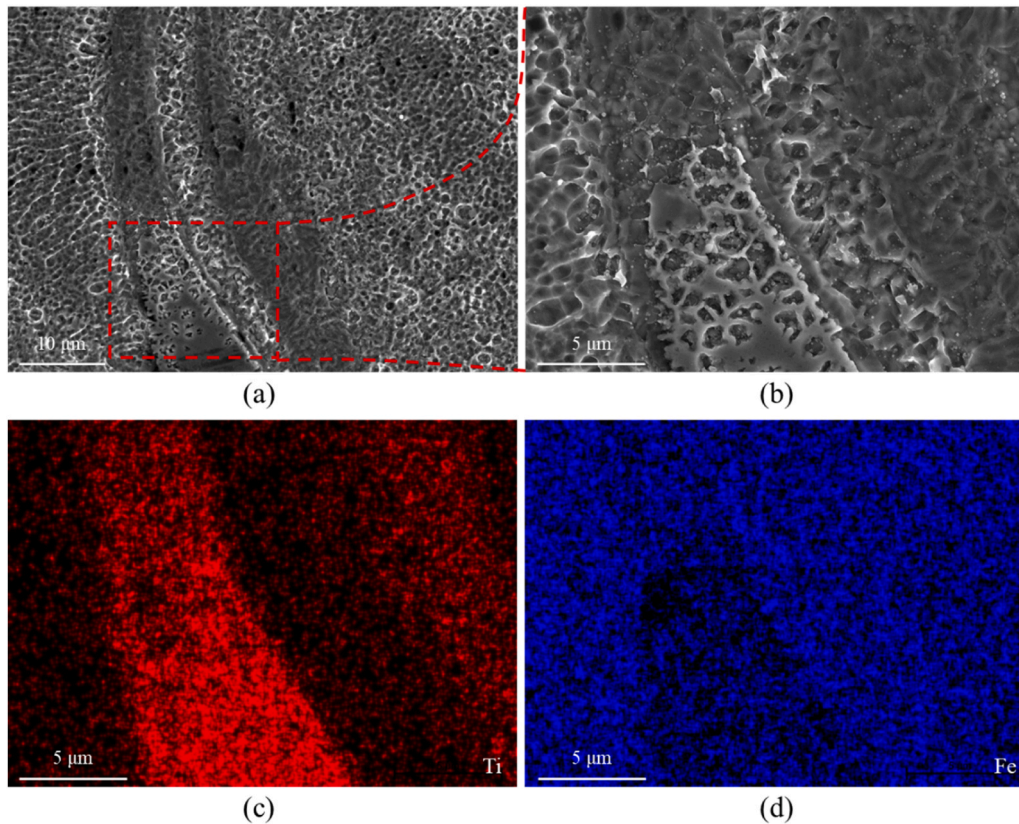


Fig. 14. Microstructures and element distribution characterize in the welding seam cross-section produced by 1# parameter. (a) microstructure, (b) enlarged picture with red box, (c) energy spectrum mapping of the Ti element, (d) energy spectrum mapping of the Fe element.

parameter 3# is approximately 20.6 N. For the joint welded with parameter 4#, the average load is about 17.6 N, and for the joint welded with parameter 5#, it is approximately 17.24 N. The joints obtained with parameters 1# and 5# show relatively stable performance, while those with parameters 3# and 4# exhibit significant performance variability. The joint welded with parameter 1# has the highest maximum average load at failure, approximately double that of the joints welded with parameters 3#-5#.

4. Discussion

4.1. Effect of fluid flow on Ti element distribution

To demonstrate the physical principles governing the metal mixing behaviour in the weld seam under both penetration and non-penetration welding conditions, the simulation results for process parameters 1# and 5# are presented below. The dynamic distribution characteristics of metal mixing in the weld pool were obtained by slicing along the cross section of the weld. Fig. 8a-d show four snapshots of the simulation process for parameter 1#, while Fig. 8e-h display four snapshots for parameter 5#. The fluid velocity vector (direction and speed) is visualized by black arrows.

For the simulation of welding parameters 1#, at time $T = t_0$, under the stirring action of the keyhole and recoil pressure, the molten metal in the weld cross-section primarily flows towards the bottom and sides of the molten pool. At time $T = t_0 + 0.0005$ s, the upward flow trend of the molten stainless steel is enhanced, carrying some Ti elements to the top of the molten pool, but there is no significant formation of a Ti-rich zone on both sides of the molten pool. At time $T = t_0 + 0.001$ s, the flow trend of the molten metal is entirely upward, resulting in the formation of Ti-rich zones on both sides of the molten pool. At time $T = t_0 + 0.0015$ s, most areas have ceased to flow, with only the molten metal at the center

of the molten pool continuing to exhibit an upward flow trend, although the flow of the molten pool has little effect on the Ti-rich zones.

For the simulation of welding parameters 5#, at time $T = t_0$, under the stirring action of the keyhole and recoil pressure, the molten metal in the weld cross-section primarily flows towards the bottom and sides of the molten pool, with a relatively uniform mixture of Ti elements throughout the molten pool. At time $T = t_0 + 0.0005$ s, the molten metal mainly flows towards the top and sides of the molten pool, with a large amount of Ti elements moving from the bottom to the top. Upon contact with the molten pool boundaries, the melt quickly solidifies, resulting in the formation of a Ti-rich zone within the molten pool. At time $T = t_0 + 0.001$ s, the upward flow trend of the molten metal is enhanced, although the bottom of the molten pool still maintains a left-right flow trend. Large amount of Ti elements moves upward from the bottom and accumulates on both sides of the molten pool. At time $T = t_0 + 0.0015$ s, the bottom of the molten pool still retains the upward and left-right flow trends, further promoting the accumulation of Ti elements towards the top and sides of the molten pool.

Fig. 9a-d presents four snapshots of the slicing process along the longitudinal section of the weld seam during the simulation of process parameter 1#. Fig. 9e-h show four snapshots of the slicing process along the longitudinal section of the weld seam during the simulation of process parameter 5#, with fluid velocity vectors (direction and speed) visualized by black arrows. The recoil pressure at the bottom of the keyhole exerts a force that pushes the fluid away from the keyhole's base, causing the flow to deflect after encountering the molten pool boundary, resulting in backward and upward fluid motion. Additionally, the Marangoni force, arising from the temperature gradient, causes the fluid to flow from the high temperature region to the low temperature region, i.e., backward to the tail of the molten pool. Once the molten material reaches the tail of the molten pool, it turns around and moves forward.

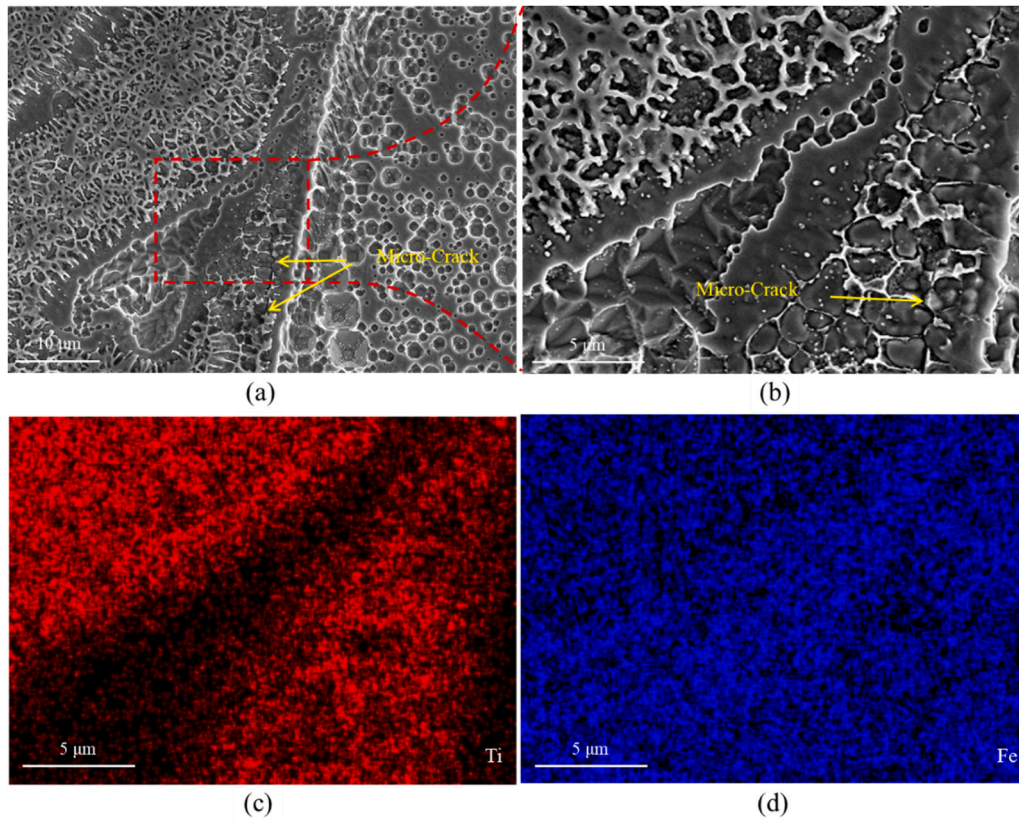


Fig. 15. Microstructures and microcracks in the welding seam cross-section produced by 4# parameter. (a) microstructure and microcracks, (b) enlarged picture with red box, (c) energy spectrum mapping of the Ti element, (d) energy spectrum mapping of the Fe element.

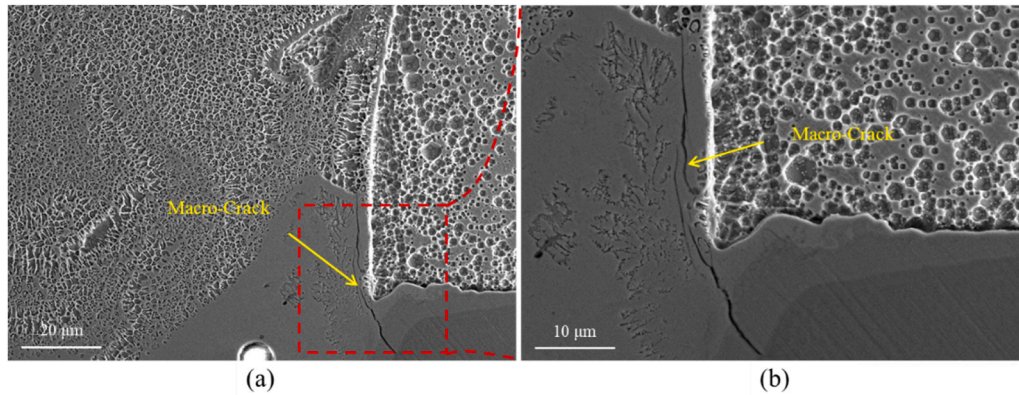


Fig. 16. Macrocracks in the welding seam cross-section produced by 4# parameter. (a) microstructure and microcracks, (b) enlarged picture with red box.

For the simulation of welding parameter 1#, there is a significant amount of Ti elements at the bottom boundary of the molten pool that have not been directly melted by the laser. These Ti elements gradually melt under the scouring action of the high-temperature stainless steel molten material and flow backward and upward with the melt. At this time, the melt immediately behind the keyhole and adjacent to it contains almost no Ti elements, as shown in Fig. 9a-b. When the melt reaches the tail of the molten pool, it begins to flow toward the keyhole, causing some Ti elements to mix into the melt adjacent to the keyhole, as shown in Fig. 9c-d. As the mixed metal solidifies at the trailing edge of the molten pool, it results in the formation of regions with high titanium (Ti) concentration and regions with low titanium concentration.

For the simulation of welding parameter 5#, the Ti elements at the bottom are heated and melted, and under the influence of the recoil

pressure and Marangoni force, the Ti elements flow backward and upward. The Fe elements at the top of the molten pool melt and flow down to compensate for the vacancy of Ti elements at the bottom, resulting in an increased Fe content at the bottom, as shown in Fig. 9e-f. Due to the lack of boundary constraints at the bottom of the molten pool, the tendency for the melt to flow backward and the flow trajectory significantly increase. The molten pool produces large flow vortices, which facilitate the mixing of Ti and Fe elements. At the same time, the backflow of Ti elements increases the Ti content at the bottom of the molten pool, as shown in Fig. 9g-h. Moreover, clusters rich in Ti can be found in the flowing area of the molten pool, indicating insufficient mixing of Ti and Fe at certain positions during the welding process. When Ti flows into the molten pool boundary, the Ti-rich fluid clusters solidify quickly before mixing with Fe.

4.2. Effect of fluid density on Ti element distribution

In this welding CFD model, the Ti thin plate is placed at the bottom of the stainless steel thin plate. Simulation and experimental results indicate that the Fe element content at the bottom of the molten pool is higher than that of the Ti element. To investigate the influence of differences in melt density on element distribution, we replaced the material properties of the Ti thin plate with stainless steel based on the Ti-stainless steel welding model. Fig. 10 shows snapshots of the simulation process along the transverse and longitudinal sections of the weld seam during the simulation of process parameter 1#. Fig. 10a-d are transverse section snapshots and Fig. 10e-f are longitudinal section snapshots.

Comparing Figs. 8a-b and 10a-b, it can be seen that the distribution of the bottom alloy at the initial melting stage is similar in the cross-section of the weld seam. Comparing Figs. 8c and 10c shows that when the densities of the melt are the same, the trend of the bottom melt flowing towards both sides of the weld seam increases. When the densities of the melt are the same, the alloy elements from the bottom are more likely to flow towards the center of the weld seam, resulting in a more thorough mixing of elements, which reduces the width of the element deficiency zone in the center of the weld seam, as shown by comparing Figs. 8d and 10d. From the element distribution maps of the longitudinal section of the weld seam, it can be seen that when the melt densities are the same, the bottom melt still exhibits an alternating distribution of high and low concentration bands within the weld. However, the overall distribution of the melt becomes more uniform, as seen in the comparison of Figs. 8e-h and 10e-h. In the stainless steel-Ti welding model, the density of the bottom melt is much lower than that of the top fluid. The bottom melt is driven not only by recoil pressure and Marangoni forces but also by buoyancy, leading to a stronger upward flow and a greater resistance to disturbance, making it less likely to flow towards the center of the molten pool. When the density of the bottom melt is nearly equal to that of the top fluid, the upward flow of the bottom melt lacks buoyancy driving force, and its ability to resist disturbance during fluid flow is weakened, resulting in an enhanced tendency to flow towards the center of the molten pool. Therefore, we can infer that in the stainless steel-Ti welded joint, the higher content of Fe elements in the Ti side of the weld seam is attributed not only to the influence of fluid flow but also to the differences in melt density.

4.3. Effect of element distribution on microstructure and mechanical properties of joint

Fig. 11a compares the content of Ti elements in the welds obtained from different welding parameters and the corresponding maximum load at which the welds failed. When counting the content of Ti elements in the welds, only those mesh elements where the Ti content is greater than “0” and less than “100 at%” are included in the statistical range. The impact of laser power on the average content of Ti element is the highest, while the impact of frequency is relatively small. As the frequency increases, the content of Ti element in the weld gradually increases. The statistical results show that during welding with process 1#, the average content of Ti in the weld is approximately 9.99 at%, with a maximum average load at failure of about 40.87 ± 2.97 N ($n = 3$). During welding with process 2#, the average Ti content is approximately 18.43 %, with a maximum average load at failure of about 30.59 ± 7.5 N ($n = 3$). During welding with process 3#, the average Ti content is approximately 23.41 at%, with a maximum average load at failure of about 20.6 ± 4.76 N ($n = 3$). During welding with process 4#, the average Ti content is approximately 25.67 at%, with a maximum average load at failure of about 17.6 ± 10 N ($n = 3$). During welding with process 5#, the average Ti content is approximately 27.54 at%, with a maximum average load at failure of about 17.24 ± 0.72 N ($n = 3$).

Fe-Ti phase diagram is shown in Fig. 11b [45]. When Ti is the main

element, the solubility of Fe in α -Ti is very low (0.1at%). When the composition exceeds this value, FeTi phase begins to form. When Fe is the main element, the maximum solubility of Ti in α -Fe is about 10 at%, but as the temperature decreases, the solubility of Ti gradually approaches zero, leading to the precipitation of the Fe_2Ti phase. The presence of these phases reduces the plasticity of the weld, and the chemical inhomogeneity in the melted zone can lead to the local accumulation of brittle phases [46]. According to the phase diagram, when Fe is the main element, the maximum solid solubility of Ti in Fe is about 10 at%. During the welding process of method #1, the content of Ti in the weld is relatively low, resulting in a lower tendency to form brittle intermetallic compounds in the weld. Additionally, simulation results indicate that during nanosecond laser welding, the max cooling rate of the weld is approximately 6.7×10^7 °C/s. This ultra-high cooling rate may retain the supersaturated α -Fe down to room temperature, thus suppressing the formation of brittle intermetallic compounds. Tomashchuk et al.[47] found that during laser welding of 316 L/V/Ti6Al6V joints, high cooling rates can have a positive effect on inhibiting the formation of brittle phase. Their research results indicated that the melted zone between the 316 L stainless steel and the V interlayer consists of a solid solution, and they inferred that the reason for this phenomenon is the rapid quenching of the melt to low temperatures, leading to the separation of the initial solid solution into V-rich and V-poor clusters. For processes 2#, 3#, and 4#, the content of Ti in the weld exceeds the eutectic point, making it difficult to suppress the eutectic reaction during solidification through rapid cooling, resulting in the formation of the Fe_2Ti intermetallic compound.

Fig. 12 displays the Electron Back-Scattered Diffraction (EBSD) image of the cross-section of the welding seam produced by 1# parameter. There is no obvious heat affected zone in the weld seam, and the grain size is very fine. The band contrast image clearly shows the striped distribution characterize of some grains, and the phase distribution image shows that the Ti element tends to aggregate towards both sides of the weld, while the stainless steel tends to aggregate towards the bottom of the melt pool. These are similar to the CFD simulation results. Furthermore, no Ti-Fe intermetallic compounds were detected in the weld seam through EBSD. Fig. 13a-d show the fracture morphology of the welded joint with process parameters 1#-5#, respectively. It can be seen from the figure that there are obvious ductile dimples in the fracture of the welded joint with parameters 1# and 2#, exhibiting a mixed fracture characteristic of toughness and brittleness. The fracture morphology of the 3#-5# parameter welded joints shows brittle fracture characteristics.

Fig. 14 shows microstructures in the cross-section of the weld seam with 1# process parameter, while Fig. 15 presents the microstructure and microcracks in the cross-section of the weld seam with 4# process parameter. Consistent with the simulation results, the distribution of Ti elements in the weld seam features alternating bands of high Ti concentration and low Ti concentration, as illustrated in Figs. 14c and 15c. Figs. 14b and 15b indicate that in the high Ti concentration band, a large number of network-like structures appear, resembling patterns formed by fluid flow. These network structures may result from the rapid cooling of the Ti melt during its flow in the molten pool. Since the melting point of Ti is higher than that of Fe, the high Ti concentration melt solidifies before the low Ti concentration melt, retaining the characteristics of flow due to rapid solidification. Additionally, in the cross-section of the weld seam with 4# process parameter, microcracks propagation along the network structure were observed in the high Ti concentration band. However, no microcracks were found in the cross-section of the weld seam with 1# process parameter. Fig. 16 presents macrocracks in the cross-section of the weld seam with 4# process parameter. Macrocracks are easily generated at the Ti/steel interface, propagation along the Ti-rich areas at the edges of the molten pool. Another significant challenge in dissimilar materials welding is the high residual stress. The thermal expansion coefficient of Ti is approximately $7.6 \text{ mm}^{-1} \text{ K}^{-1}$, while that of stainless steel is about $17\text{--}18 \text{ mm}^{-1} \text{ K}^{-1}$,

which differs by nearly a factor of two. This discrepancy leads to residual stress during solidification of the molten pool, potentially causing interface cracking. Hao et al. [48,49] pointed out that cracks are likely to propagate in the transition zone of Ti/steel welds, limiting the tensile strength of the resultant joints. The presence of microcracks and macrocracks in the cross-section of the weld seam with 4# process parameter results in lower weld strength. In contrast, the absence of both microcracks and macrocracks in the cross-section of the weld seam with 1# process parameter, which can significantly enhance weld strength.

Due to the complexity of residual stress at the interface, several factors are involved, including differences in physical properties of dissimilar metals, the extreme thermal environment of the molten pool, and the distribution state of dissimilar metal elements. Currently, there is limited research on residual stress in dissimilar metal welded seams [50,51]. Future work can investigate the influence of differences in metal physical properties, the extreme thermal environment of the molten pool, and the distribution state of dissimilar metal elements on residual stress through a coupling of finite element method (FEM) and computational fluid dynamics (CFD).

5. Conclusion

This study develops a three-dimensional Computational Fluid Dynamics (CFD) model for the nanosecond laser welding of thin-walled dissimilar metals, specifically pure titanium and 304 stainless steel, and validates the model using experimental results. The model elucidates the distribution patterns of titanium elements in the molten pool under different flow modes and reveals the factors affecting the uneven distribution of titanium elements. It quantitatively analyzes the impact mechanism of titanium element content in the weld on the performance of Ti-stainless steel joints. A method has been proposed to inhibit the formation of brittle intermetallic compounds during the welding of thin-walled dissimilar metal materials, which holds promise for application in precision structural welding fields such as aerospace, medical devices, and consumer electronics. The main conclusions are as follows:

- (1) During the nanosecond laser welding of titanium-stainless steel dissimilar metals, regardless of whether the molten pool penetrates the titanium thin wall, titanium elements tend to form enrichment zones at the edge of the molten pool on the stainless steel side. This formation of the Ti enrichment zone is primarily due to the rapid solidification that occurs after the stainless steel melt carrying Ti melt makes contact with the boundary of the molten pool as it flows upward. Additionally, the lighter density of titanium leads to a stronger upward flow tendency of the Ti melt, while the tendency to flow laterally is weaker.
- (2) The distribution of titanium elements in the weld exhibits alternating bands of high Ti concentration and low Ti concentration. Within the bands of high Ti concentration, numerous network structures appear, resembling patterns formed by fluid flow. These network structures may preserve the characteristics of the flow during the rapid solidification of the rich Ti melt. Microcracks in the weld are prone to propagate along the edges of the network structures within the high Ti concentration bands, while macro-cracks are likely to occur at the Ti/steel interface.
- (3) The content of Ti in the weld seam is 9.99 %, no intermetallic compounds or micro cracks have occurred. At the same time, the strength of the weld can be increased by about twice. This study provides a new option for welding thin-walled titanium-stainless steel dissimilar alloys, which is to achieve welding by adjusting the proportion of dissimilar elements in the weld seam, rather than adding additional alloys or brazing materials.
- (4) Adjusting the frequency can fine tune the proportion of dissimilar elements in the weld seam. As the frequency increases, the content of Ti element in the weld seam gradually increases.

Subsequent research can fine tune the proportion of dissimilar elements in the weld seam to further optimize the performance of the joint.

CRediT authorship contribution statement

Qiao Rui-Hua: Data curation. **Zhang Yu-Ruo:** Validation. **Wang Xiang:** Writing – review & editing, Writing – original draft, Software. **Ma Jun:** Methodology. **Zhang Lin-jie:** Supervision, Conceptualization. **Liang Che-Ping:** Investigation. **Suck-Joo Na:** Supervision.

Declaration of Competing Interest

The authors declared that they have no conflict of interest to this work.

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